

TASK 1 — Behave BDD

1. Experiment Title

Implementation of Selenium Python and Behave BDD

2. Problem Statement

Implement a web automation test using **Selenium with Python and Behave BDD** to verify the login functionality of a web application using a Gherkin feature file and Python step definitions.

3. Objective

- To set up and use the **Behave BDD framework** with Python.
- To create test scenarios using **Gherkin syntax**.
- To automate the login functionality using Selenium WebDriver.
- To execute and verify an end-to-end login scenario.

4. Tools, Software and Concepts Used

Tools & Software

- Python
- Behave
- Selenium WebDriver
- Google Chrome
- PyCharm / VS Code

Concepts Used

- Behaviour-Driven Development (BDD)
- Gherkin
- Feature
- Scenario
- Given
- When

- Then
- Selenium WebDriver
- Web Element Locators
- Assertions

5. Source Code / Implementation Steps

Project Structure

Assignment_1_Behave_BDD

```
|
└─ features
    |
    └─ login.feature
        |
        └─ steps
            └─ login_steps.py
```

login.feature

Feature: Login functionality

Scenario: Successful login with valid credentials

Given I open the SauceDemo login page

When I enter valid username and password

And I click the login button

Then I should be logged in successfully

login_steps.py

```
from behave import given, when, then
```

```
from selenium import webdriver
```

```
from selenium.webdriver.common.by import By
```

```

@given("I open the SauceDemo login page")
def step_open_login_page(context):
    context.driver = webdriver.Chrome()
    context.driver.maximize_window()
    context.driver.get("https://www.saucedemo.com/")

@when("I enter valid username and password")
def step_enter_credentials(context):
    context.driver.find_element(By.ID, "user-name").send_keys("standard_user")
    context.driver.find_element(By.NAME,
"password").send_keys("secret_sauce")

@when("I click the login button")
def step_click_login(context):
    context.driver.find_element(
        By.XPATH, "//input[@type='submit']"
    ).click()

@then("I should be logged in successfully")
def step_verify_login(context):
    assert "/inventory.html" in context.driver.current_url
    print("Login successful!")
    context.driver.quit()

```

Execution

From the Assignment_1_Behave_BDD folder, run:

```
behave
```

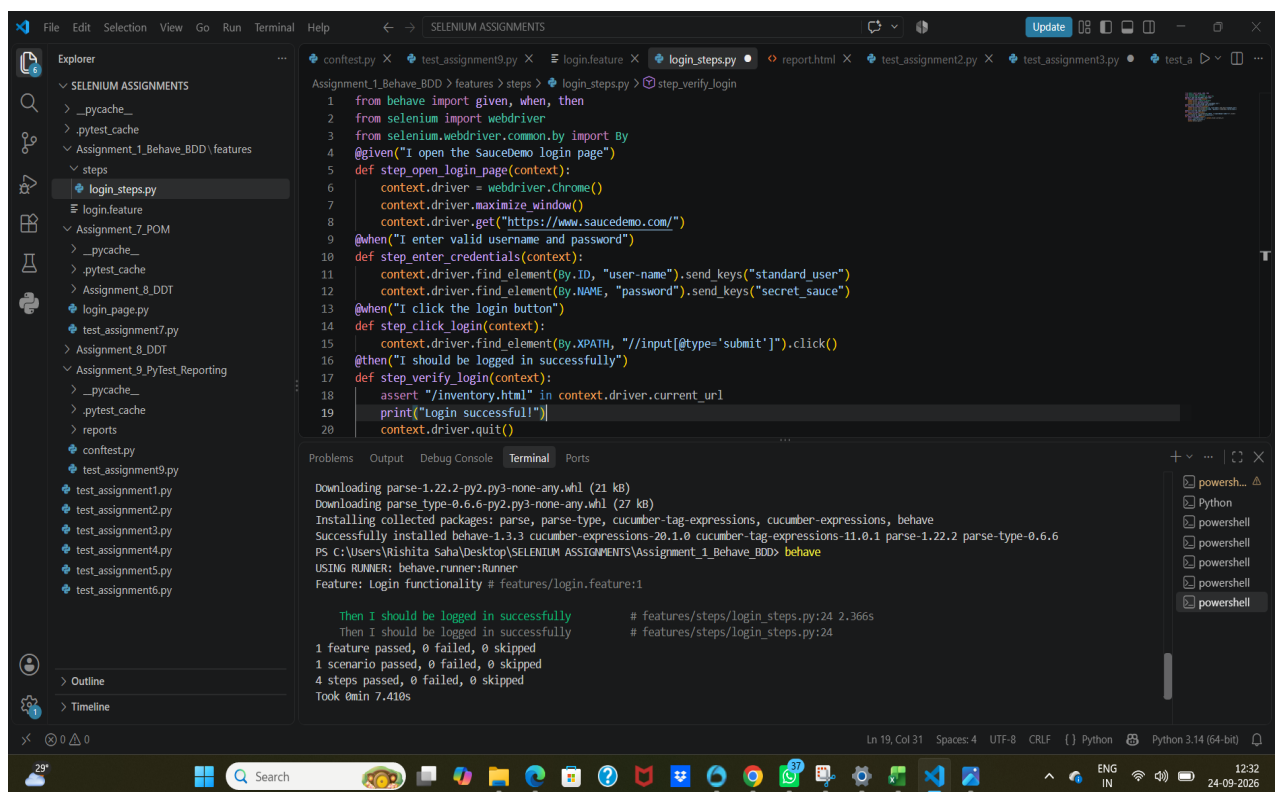
The successful execution should show:

1 feature passed, 0 failed, 0 skipped

1 scenario passed, 0 failed, 0 skipped

4 steps passed, 0 failed, 0 skipped

6. Output with Clearly Labelled Screenshots



Show the terminal after:

behave

with the feature, scenario, and four steps passed.

7. Result

The Selenium-based login scenario was successfully implemented using **Python and Behave BDD**. The feature file was executed successfully and all four steps passed.

8. Observation

- Gherkin provides a readable structure for describing test scenarios.
- Given, When, and Then steps are mapped to Python functions.
- Selenium WebDriver performs the actual browser automation.
- The successful login was verified by checking the inventory.html URL.

9. Conclusion

This experiment demonstrated the implementation of **Selenium Python automation using the Behave BDD framework**. The Gherkin feature and Python step definitions were successfully integrated to execute an end-to-end login test.